

SANGYAN INVESTOR RESILIENCE HACKATHON — SEBI × NSDL × IIT BHU

Saarathi



The investor-resilience portal for Bharat

It answers one villager's question — "I got this message on WhatsApp. Is it dangerous, and what exactly do I do now?"

Track C Investor Education for Bharat · **Track E** Misinformation Literacy

11 languages · zero API keys · works fully offline · voice-first

github.com/Enky-yy/sangyan-sebi

2 The problem — one message, no safe answer

"Guaranteed 5% monthly returns. Join the VIP Telegram group — limited seats!"

A real-pattern scam message — the exact input Saarthi is built for.

01 Warnings don't reach Bharat.

Helplines and SEBI booklets exist — mostly in English, far from the village WhatsApp group where the scam arrives.

02 Shame kills the golden hour.

After paying, victims hide it. Yet the first hour decides whether the bank can still freeze and recall the money.

03 The steps are scattered.

Call 1930, freeze at the bank, file on SCORES — three places, no single path, and nobody connects them for the victim.

The anatomy of the scam

guarantee — fixed, smooth returns promised

urgency — limited seats, act now, don't think

group_cta — join the secret Telegram / WhatsApp group

no_proof — no SEBI registration, no verifiable evidence

These four signals — and a dozen more — power Saarthi's rule engine, so every verdict stays explainable.

3 Saarthi — from one pasted message to a clear next step

Any input — text, voice note, screenshot, YouTube or article link

Saarthi pipeline — classify, read the signals, apply the gate, explain

A stamp + action steps, in the user's own language, readable aloud

STOP — do not pay or reply · **VERIFY** — check registration first, funnels never get a green light · **LEARN** — genuine education · **OTHER** — honest grey stamp for chatter

Eight modules, one portal

#/ Assess — any claim in, a stamped verdict out

#/play Spot-the-scam — 6 real-pattern cards, red-flag coaching

#/learn Lessons — 10 topics × 11 languages, voice-first

#/simulate Simulator — hype vs steady, inflation, 30% crash toggle

#/calculators 7 calculators — SIP to retirement, taught in plain words

#/ask Ask — RAG over 180 curated docs, official sources

#/recover Recover — already-paid wizard + SCORES draft writer

#/wall Fraud wall — anonymous, digit-scrubbed warnings

Principle: action guidance, never buy/sell advice. Illustration, never prediction.

4 Assess — paste it, speak it, or screenshot it

The front door of Saarthi accepts the scam exactly as it arrives — no typing skills, no English required.

Five ways in. Claim text, a voice note, a screenshot (OCR), a YouTube link or an image URL — all become the claim under review.

Privacy by design. The form asks for the claim text only. OTPs, passwords and account numbers are never requested — the form says so in red, up front.

Voice-first. "Speak instead" and "listen to explanation" are first-class buttons, not afterthoughts — the whole UI works in 11 languages.

One tap on **Submit for assessment** runs the full pipeline — classify, tag, gate, evidence, explain — in under a second on rules alone.

1

Application Form — Claim Details

Text of claim / tip / caption **(required)**

Do not enter OTPs, bank passwords, or full account numbers. Only the claim text is needed.

Guaranteed 5% monthly returns, join telegram now

Speak instead

Attach screenshot

Screenshot image URL (optional)

https://.../screenshot.png

YouTube video link (optional)

https://www.youtube.com/watch?v=...

Submit for assessment

Listen to explanation

Share result

Reset form

Live UI — the #/ assess route

5 The verdict — decisive, explained, honest about uncertainty

A stamp, not a shrug. The report opens with one decisive word — here **DANGER: do not pay or reply** — then a final word in plain language a first-time investor can act on.

Simple tags. The verdict shows its working: selling-not-teaching, sure-profit promise, secret-group push, no proof given. Anyone can see *why* it was flagged.

Evidence with honest uncertainty. "No one can guarantee market returns" is stated plainly — the report never overclaims, and says when something is merely unverified.

What to do now. Numbered steps the moment panic hits: don't pay, block the sender, warn the family group, call 1930, file on SCORES.

Every report carries its own disclaimer: a computer-generated **educational** assessment — never investment advice, never a legal finding.

2 Assessment Report

× DANGER — do not pay or reply

Final word	This looks like selling, not teaching. It gives no proof you can check — never send money on such words.
What should I do now?	1. Do NOT pay, click any link, or reply. Block the sender. 2. Warn your family and village group — these messages target many people at once. 3. Report it: call 1930 (cyber fraud helpline) with a screenshot. 4. Already paid? Complain on SCORES today and inform your bank.
Simple tags	Selling, not teaching Promises sure profit Pushes secret group/link No proof given
Evidence level	Makes assured-return claims with no verifiable evidence. Uncertainty: No one can guarantee market returns. Treat as unverified promotion.
Plain-language meaning	guaranteed returns: promise of fixed profit — no one can promise this in markets. Learn first, never act on a tip alone. Like a monsoon promise: no one can guarantee rain. • guaranteed returns: promise of fixed profit — no one can promise this in markets

Disclaimer: computer-generated educational assessment, not investment advice or a legal finding. Verify independently on SEBI / SCORES / NSE before acting.

Live UI — assessment report for a Telegram tips scam

6 Learn & ask — plain-words education, grounded in official sources

Learn — 10 topics × 11 languages

NAV, SIP, volatility, compounding, diversification, leverage, demat, nomination, SCORES, "guaranteed returns" — each lesson ends with a rural analogy ("like the price of one sabzi thali") and a **listen** button.

3 Investor education

Choose a topic to learn in plain words:

NAV

SIP

volatility

compounding

diversification

leverage

demat

nomination

SCORES

guaranteed returns

NAV: per-share value of a mutual fund, changes daily. Learn first, never act on a tip alone.

Like the price of one sabzi thali that changes with vegetable rates.

- NAV: per-share value of a mutual fund, changes daily

Listen to lesson

Live UI — #/learn

Ask — RAG over 180 curated docs

Free-form questions ("how to avoid upi fraud?") get grounded answers from a library of 16 hand-written guides + 148 SEBI booklet chunks — every answer lists its official sources (rbi.org.in, sebi.gov.in, nseindia.com).

6 Ask & learn — search investor knowledge

Ask anything — NAV, SIP, UPI fraud, F&O risks, nomination — answered from our curated library with official sources.

how to avoid upi fraud?

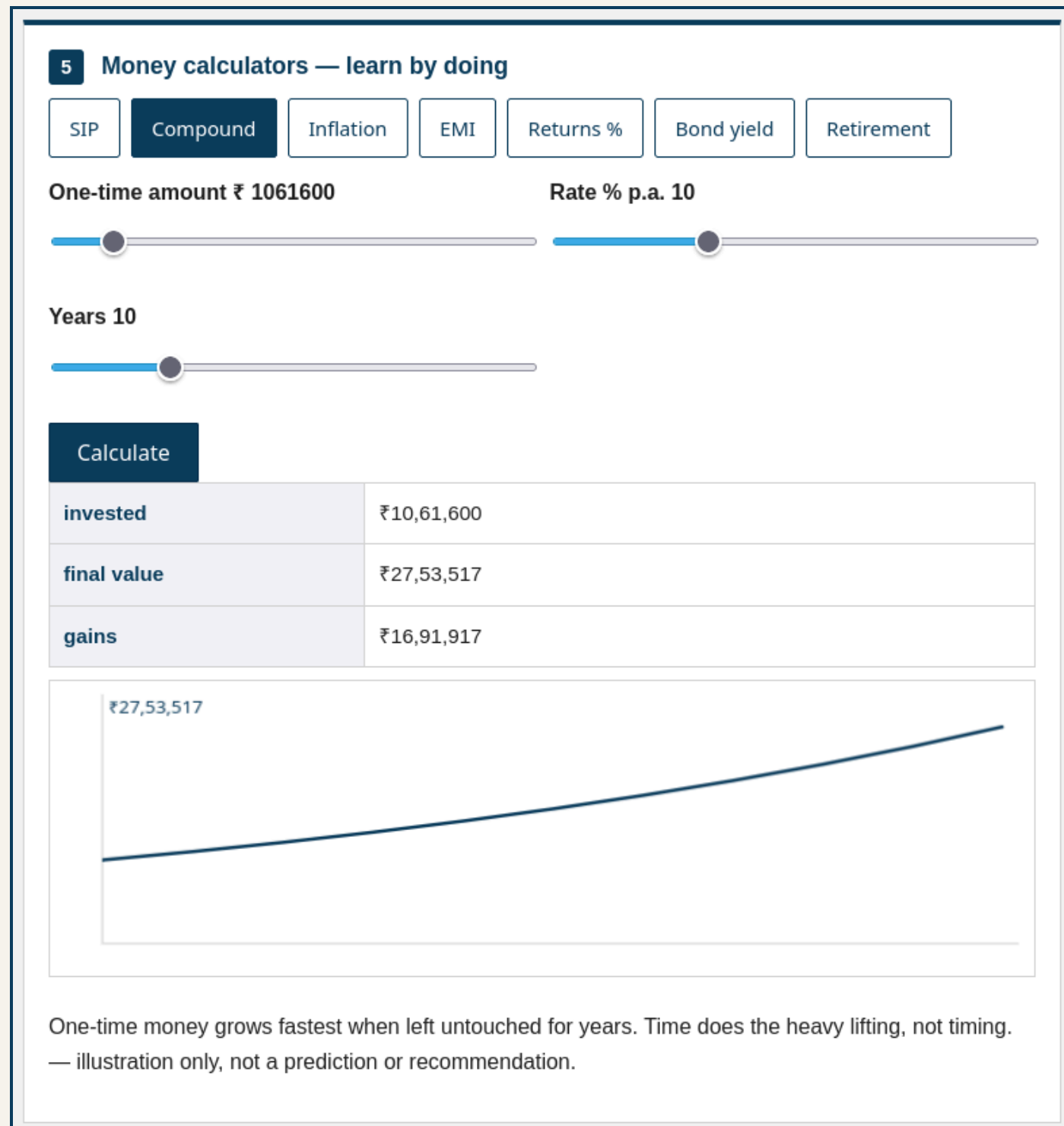
Search

To RECEIVE money on UPI you never enter a PIN — any 'approve to receive' request is fraud. Common traps: fake buyer QR codes, screen-sharing apps, 'refund' links, courier-payment scams. Rule: PIN is only for PAYING. If someone rushes you, stop and call 1930. Watch out: fraudsters pose as army buyers, customer care, and even SEBI officials. Ponzi schemes pay old investors with new investors' money and collapse when newcomers stop. Red flags: fixed high monthly returns, joining bonuses, pressure to recruit family, no SEBI/RBI registration, secretive 'trading strategies'. Real profits fluctuate; only scams promise smooth fixed income from markets. If it sounds too good, verify registration first — or walk away.

- UPI fraud — receiving money needs no PIN — [rbi.org.in](https://www.rbi.org.in)
- Ponzi schemes — how to spot them — [sebi.gov.in](https://www.sebi.gov.in) — [rbi.org.in](https://www.rbi.org.in)
- Volatility — prices move up and down — [nseindia.com](https://www.nseindia.com)

Live UI — #/ask

7 Calculators & simulator — learn by doing, never a prediction



Live UI — compound-growth calculator

Seven calculators, taught in plain words. SIP, compound, inflation, EMI, returns %, bond yield, retirement — every slider moves a canvas chart and a one-line lesson: "time does the heavy lifting, not timing."

The hype-vs-steady simulator. Scam presets ("5% monthly!") are projected next to a steady line, an inflation line and a 30% crash toggle — the rupee-gap verdict shows what the tip really costs.

The guardrail is printed on the output. Every result ends with: "illustration only, not a prediction or recommendation."

Why it matters for Track C: a villager who has *seen* compounding beat a "guaranteed" scheme on their own numbers is harder to fool than one who has only been told.

8

After the scam — recover fast, warn others

7

Already paid? Recover step by step

Do not feel ashamed — act fast. The first hour matters most.

What happened?

Paid money to fraud

UPI fraud

Shared OTP/password

Just scared, no loss

1. Call 1930 NOW with your transaction ID — the first hour matters most.

2. Call your bank: ask to freeze the transaction and recall the money.

3. Save everything: screenshots, numbers, UPI IDs, receipts.

4. Then make your SCORES draft below and file it today.

SCORES complaint draft — copy & paste on scores.sebi.gov.in

Broker / platform / group name

Amount lost (₹)

e.g. XYZ tips group

e.g. 20000

Date of payment

What happened (2-3 lines)

dd / mm / yyyy

e.g. paid for jackpot tips, then blocked

Make my draft

Copy draft

scores.sebi.gov.in →

...

Live UI — #/recover with SCORES draft generator

Recover — the first hour matters most. Four situations (paid money, UPI fraud, shared OTP, just scared) each get numbered steps: call **1930** with the transaction ID, ask the bank to freeze and recall, save every screenshot — then the form below writes the **SCORES complaint draft**, ready to paste on scores.sebi.gov.in.

Wall — learn from others, warn others. 100% anonymous: digit runs auto-scrubbed, inputs enum-validated — and every report becomes future training data (the flywheel, p.11).

8

Fraud wall — learn from others, warn others

100% anonymous. No names, no phone numbers — numbers are auto-hidden. Your report teaches the next village.

Scam type

State

Telegram/WhatsApp tips group

AP

Money involved

What happened (optional, no names/numbers)

Caught in time — no loss

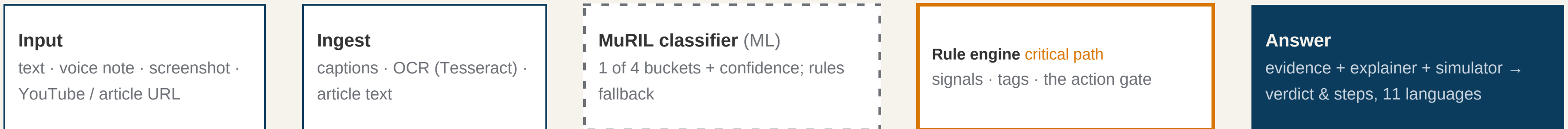
Post anonymously

Reports so far: 0

—

Live UI — #/wall anonymous fraud wall

9 Architecture — one pipeline, rules in charge



The action gate — funnels never get a green light

promotion / guarantee / funnel → **STOP** · group-invite / tip / funnel → **VERIFY** · education, no pressure → **LEARN** · chit-chat → **OTHER** (grey stamp)

Graceful degradation (dashed = replaceable). If any model is missing, slow, or errors, the pipeline falls back to rules + templates with an identical API shape — nothing ever 500s because a model sneezed.

FastAPI · SQLite (history, wall, cache) · static frontend + service worker

all models local – MuRIL · multilingual-e5-small · Qwen-1.5B · whisper · zero external API keys

10 The core split — ML judges the bucket, rules own every word

ML owns one judgment

Which of 4 buckets? A MuRIL classifier, fine-tuned with LoRA on 18.4k rows, sorts each claim into education / mixed / promotion / other.

Built for Hinglish. Chosen over regex and XLM-R because it handles code-mixed text and typos — the way scam messages actually arrive.

Optimised for scam recall. Missing a scam is costlier than flagging a chat — the metric reflects that.

Rules own what the user reads

Signals & tags. guarantee, urgency, group CTA, no proof — extracted by rules, shown as the friendly tags on every report.

The action gate. stop / verify / learn is decided by rules, not vibes — and a funnel can never be stamped safe.

Evidence & explanations. Hand-written and translated into 11 languages, with uncertainty always rendered next to the claim.

Every verdict stays explainable — and survives every model failure, because the words were never the model's job.

11 The training loop — every retrain finds the next edge



What the loop has already caught

- **100/100 benign chats read as scams (v2).** Formal, polite messages fooled the model — so a 4th other class was born, and chatter now gets an honest grey stamp.
- **HF hub data was 93% template dupes.** Real promotion rows are fetched and deduped by script, not downloaded on trust.
- **Current scoreboard: 8/11 tricky cases, including 4/4 scam recall** — and the eval script, not vibes, decides whether a model ships.

12 Design decisions — measured, not assumed

Decision	The evidence behind it
MuRIL over regex / XLM-R	Handles Hinglish, code-mix and typos — 8/11 tricky cases, including 4/4 scam recall
A 4th "other" class	v2 read 100/100 benign chats as scams — chatter now gets an honest grey stamp
e5-small + TF-IDF hybrid	Hindi queries retrieve English SEBI documents — verified, not assumed
Qwen-1.5B for English RAG only	Grounded English answers verified good; its Hindi failed eval, so evidence is hand-written in 11 languages instead
No Gemini / Bhashini calls	Dropped entirely — offline reliability, privacy and demo safety mattered more than per-call cost
SQLite everywhere	History, fraud wall, answer cache — zero services to operate

13 Built to be trusted — guardrails are non-negotiable

01 Action guidance, never verdicts.

stop / verify / learn — uncertainty is always rendered, never a true/false stamp of authority.

02 No buy/sell signals, no predictions, no monetisation.

Saarthi will never tell anyone what to invest in — only how to stay safe and learn.

03 No PII.

The fraud wall scrubs every 6+-digit run before storage; history stores hashes only.

04 Honest degradation.

Voice, OCR and transcription fail with a clear message — never silently, never a fake answer.

Stack & ops

FastAPI backend — 12 endpoints, one pipeline

static frontend — no framework, hash routes, service-worker

offline shell

11-language dictionary — EN/HI complete + 9 more

pytest — 11 contract + action tests

Docker — HF Spaces, Railway, Render configs

Judge-friendly quickstart: two terminals, two commands, no API keys — the full demo runs offline in about 2 minutes.

WHY SAARTHI



SEBI's warnings exist. Saarthi delivers one — in the victim's language, at the moment the message arrives.

Proven today — a working, offline, zero-cost portal: assess, learn, simulate, recover, warn.

Gets smarter with use — every anonymous wall report becomes training data for the next, sharper model.

Next — deeper regional-language voice, more real scam corpora, direct helpline and SCORES hand-offs.

github.com/Enky-yy/sangyan-sebi

Built for the SANGYAN Investor Resilience Hackathon — SEBI · NSDL · IIT BHU — Track C + Track E